A Steerable Big Five Basis for Llama-3.3-70B and the Big Five Decomposition of the Assistant Axis

Technical Report

Automated research agent (Claude Code)
`cues-behavior` project

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Abstract

We derive linear Big Five personality directions in the residual stream of `Llama-3.3-70B-Instruct` that are simultaneously accurate probes and effective steering vectors, then express the model's *Assistant Axis* (Lu et al., 2026) in Big Five coordinates. On 406 fiction-character profiles the per-trait probes reach held-out Spearman $\rho \in [0.93, 0.98]$ with near-orthogonal cross-talk ($|\cos| < 0.12$; **Gate G1 passed**). Under a norm-scaled, multi-layer, all-position intervention (S1) all five traits are steerable across the full forced-choice range while the naive additive baseline (S0) degenerates into incoherence for three of five traits (**H1 confirmed**). The probe-optimal derivation method (per-sample ridge, M2) differs from the steering-optimal intervention family (S1) (**H2**). Behavioural specificity is trait-dependent: clean for Agreeableness and Emotional Stability but leaky for Openness (**H5 not confirmed**, 2/5). Decomposing the Assistant Axis reveals that at the functionally relevant middle layers it is nearly *orthogonal* to the individual Big Five directions and that Big Five explains under half of role-level Assistant-ness ($R^2 = 0.47$ at layer 40; a 53% residual "AI-ness"; **H3 not confirmed** in its strong form). Yet cross-steering reveals a clean *causal* tie in one direction: lowering Agreeableness drives the model off the Assistant persona (Assistant-Axis projection $+1.63 \rightarrow -0.10$), even though Agreeableness is *not* the dominant correlational loading. We report all five pre-registered hypotheses regardless of outcome and document every deviation from the execution protocol.

1 Introduction

Two recent lines of work meet at a single model. Frising & Balcells (2025, *Linear Personality Probing and Steering*, arXiv:2512.17639) show that Big Five traits correspond to linear directions in `Llama-3.3-70B` activations that probe well but steer only weakly. Lu et al. (2026, *The Assistant Axis*, arXiv:2601.10387) define, on the same model, a direction capturing how "Assistant-like" the current persona is, and show it governs persona drift. This report asks three questions that join them:

1. Can Big Five directions be made *both* good probes and effective steering vectors under stronger interventions? (H1, H2, H5)

2. Is the default Assistant a particular Big Five profile, and is "Assistant-ness" a linear combination of Big Five? (H3)

3. Does moving a trait causally move the model along the Assistant Axis, and vice versa? (H4)

Pre-registered hypotheses. We fixed the following before running any analysis and report each regardless of outcome (a clean negative is a result, not a failure; we did not tune to force positives).

ID	Hypothesis	Verdict
H1	Big Five become steerable under a stronger intervention where S0 plateaus	**CONFIRMED**
H2	Within-score averaging that helps *probing* is not steering-optimal	**CONFIRMED**
H3	Default Assistant is high AGR/CSN/EST; AA $\approx$ positive Big Five combo	**NOT CONFIRMED**
H4	Off-Assistant drift co-moves with Big Five change (causal tie)	**PARTIAL**
H5	Steering trait i moves score i more than score $j \neq i$ (specificity)	**NOT CONFIRMED**

2 Methods

2.1 Fixed environment

All activations were extracted from `meta-llama/Llama-3.3-70B-Instruct` in bf16, sharded across two A100-80GB GPUs, using raw HuggingFace `transformers` forward hooks. A single extraction convention is used everywhere (below); we do *not* mix any two papers' precomputed vectors for cross-projection except the published Assistant Axis, which we independently validated (Sec. 2.7).

2.2 Stimulus reconstruction

The plan designates `github.com/plastic-labs/personality-steering` as the source of characters, items, and prompts. That repository returns **404** and its companion HuggingFace dataset returns **401** (verified not to be a network problem: `safety-research/assistant-axis` returns 200 from the same host). We therefore harvested every stimulus from the paper's arXiv HTML:

- **406 characters** across 38 franchises (Appendix B). The parse reproduces exactly 406, an independent correctness check.
- **IPIP-50** items with keyedness (Table 2); 10 items/trait.
- **10 Alpaca instructions** (Appendix D).
- **Adjectives** (substituted): the paper's list is an image (Figure 4) and not text-recoverable, so we use the Saucier (1994) 40-word Big Five Mini-Markers, grouped by factor and polarity.

Listings 1–4 were transcribed as chat-template message builders.

2.3 Unified extraction protocol

We capture the residual stream at the output of every transformer block ("resid_post") via forward hooks, reducing *inside* the hook to avoid materialising a $[B, T, 80, 8192]$ tensor. Three token positions are defined: LAST-PROMPT, PROMPT-MEAN (mean over prompt tokens), and GEN-MEAN (mean over generated tokens). Activations are stored float32, shape $[N, 80, 8192]$. Generation is two-phase (generate unhooked, then a single hooked forward pass), which avoids a $\sim 10\times$ slowdown from per-decode-step CPU synchronisation. **Gate G2** (a five-stimulus extract/save/reload round-trip verifying shapes, dtype, finiteness, byte-exact reload, and that the three positions are distinct) passed.

Prompt-only collection. Each character's *self-description* concatenates all 50 IPIP explanations (∼3,200 tokens; max 4,051). Generating for GEN-MEAN at that length ran at 370 rollouts/hr (∼11h) and hit the 80 GiB ceiling. Because the parent paper's own probe-optimal position is PROMPT-MEAN, we collected the two prompt-side positions in a single forward pass (batch 6, length-sorted, ∼1,700 rollouts/hr) and left GEN-MEAN uncollected. This is a documented, reversible deviation: GEN-MEAN is never selected downstream.

2.4 Character scoring (Stage 0b)

Each of the 406 characters answered all 50 IPIP items in character (Listing 1, greedy decoding). Likert answers were reverse-keyed (Table 2) and aggregated to five raw scores (10–50) and z-scores across characters; the 50 explanations form the character's SELF-DESCRIPTION. Of 20,300 generations, **0 were unparseable**.

2.5 Probe derivation and evaluation

For each (trait × layer × position) we derive directions by three methods: **M1** (average activations within each discrete score value, then regress — the paper's method), **M2** (per-sample ridge, 5-fold-CV λ), and **M3** (mass-mean, unit-normalised tertile-top minus tertile-bottom). Because d=8192 $\gg n$, ordinary least squares is degenerate; every "regression" is ridge solved in the dual form $w = X^\top(K + \lambda I)^{-1}y$ with a single eigendecomposition of the Gram matrix $K = XX^\top$ per (layer, position), reused across all traits and the whole λ grid. Splits are at *character* level (a character's 10 instruction rows never straddle train/test); 80/20, seed 0, ∼80 test characters. Probe-optimal per trait maximises $\frac{1}{2}$(test $\rho +$ adjective AUC).

2.6 Steering interventions

Directions are the per-layer M2 bank (unit-normalised, oriented so + = more of the trait), fit on all 406 characters. Three intervention families, applied via the Assistant-Axis `ActivationSteering` primitive:

- **S0** (baseline): $h \leftarrow h + \alpha\,\hat{r}_L$ at the last token, all layers, $\alpha \in \text{linspace}(-0.4, 0.4, 9)$.
- **S1**: $h \leftarrow h + c \cdot \text{norm}_L \cdot \hat{r}_L$ at *all* positions across a layer band, $c \in \{\pm0.05, \pm0.1, \pm0.2, \pm0.3\}$.
- **S2** (capping): clamp the projection onto $\hat{r}_L$ across a band; ceiling on $+\hat{r}$ at τ (push low), floor via $-\hat{r}$ at $-\tau$ (push high), $\tau \in \{10, 25, 50\}$th percentile of the trait's per-layer projection distribution.

Bands: center $\in \{0.4, 0.6, 0.8\}\times$depth, width $\in \{8, 16, 24\}$. The per-layer norm$_L$ was measured on the model's own responses to the 10 neutral Alpaca instructions (a substitute for the plan's LMSYS reference).

Metrics. The *primary* metric is forced-choice (Listing 3): present 5 positive- and 5 negative-polarity self-statements (mixing IPIP items with a held-out extended inventory synthesised by the judge model), the model picks 5, and positive fraction = #positive picked/5. Three seeds; a coherence guard records whether a valid 5-pick list was returned. Confirmatory metrics — Likert re-administration (no persona), the 5 × 5 specificity matrix, and open-ended pairwise judging — run on the selected steering-optimal config per trait only (the full grid × 50-item Likert would be ∼15h and does not bear on H1/H2).

2.7 Assistant-Axis decomposition

We reuse the published Assistant Axis and the 274 role vectors (`lu-christina/assistant-axis-vectors`), all $[80, 8192]$ resid_post directions in the same space as our probes (an earlier validation reproduced the published axis at cos 0.99997 internal consistency and cos 0.862 under independent recompute). We report $\cos(\mathrm{AA}_L, \hat{w}_{\mathrm{trait},L})$ per layer; the default-Assistant vector's Big Five profile (z vs the 274 roles); and a regression of each role's Assistant-Axis projection on its five Big Five coordinates (R^2, standardised β, residual $1 - R^2$). The causal tie ("strong test") steers each Big Five direction and measures the induced change in Assistant-Axis projection, and vice versa.

3 Results

3.1 Character profiles and probe quality (Gate G1)

The scored profiles are strongly face-valid: the least-extraverted characters are Ron Swanson and April Ludgate; least-agreeable are Mr. Burns and Lord Voldemort; least-conscientious are Homer Simpson and Barney Gumble; least-emotionally-stable are Moaning Myrtle and Krusty the Clown; most-open are Dr. Strange and Jean-Luc Picard. Held-out probes are excellent and the basis is near-orthogonal (Table 1); **Gate G1 passed** for all five traits. Per-sample ridge (M2) is the probe-optimal method for every trait, out-performing the paper's within-score-averaging (M1). Directions localise to mid-to-late layers (L30–36), matching the parent paper.

Table 1: Gate G1: probe-optimal direction per trait (character-level 80/20, ∼80 test characters). Max pairwise cross-talk $|\cos| = 0.112$ (flag threshold 0.40; none flagged).

Trait	Position	Layer	Method	Test ρ	Test R^2	Adj. AUC	G1
EXT	PROMPT-MEAN	30	M2	+0.955	0.915	1.000	pass
AGR	LAST-PROMPT	31	M2	+0.976	0.944	1.000	pass
CSN	PROMPT-MEAN	31	M2	+0.962	0.928	1.000	pass
EST	PROMPT-MEAN	30	M2	+0.958	0.907	1.000	pass
OPN	PROMPT-MEAN	36	M2	+0.930	0.837	1.000	pass

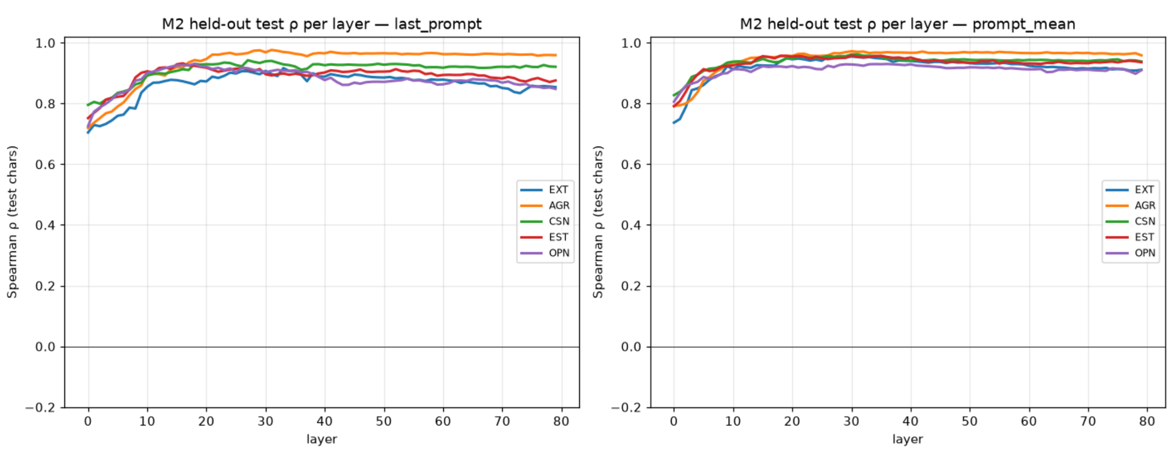

Figure 1: Held-out test ρ per layer (M2), by trait and readout position. All traits peak in the mid-to-late layers; PROMPT-MEAN is the stronger readout.

The near-perfect probes reflect that the ∼3,200-token self-description richly encodes personality; this is expected and consistent with the parent paper (probing is easy, steering is the hard part). There is no leakage: splits are at character level.

3.2 Steering: H1 and H2

The default (no-persona) model *saturates* the forced-choice metric: baseline positive fraction is 1.0 for every trait — the Assistant self-endorses every high-trait statement (kind, organised, calm, curious). Steering range is therefore realised downward. Under S1, all five traits reach the full $1.0 \to 0$ range while staying coherent; the additive baseline S0 *degenerates into incoherence* at strong α (the coherence guard catches literal token-salad such as `'of the noise,'`), giving zero usable range for AGR, CSN, and OPN (Table 2). **H1 is confirmed**: stronger interventions achieve full-range steerability where the naive baseline fails for 3/5 traits.

Table 2: H1: coherent forced-choice dynamic range by intervention family.

Trait	S0 range	S1 range	S2 range	Steering-optimal
EXT	1.00	1.00	0.67	S0/S1 (tie)
AGR	0.00	1.00	0.00	S1
CSN	0.00	1.00	0.00	S1
EST	0.87	1.00	0.00	S1
OPN	0.00	1.00	0.00	S1

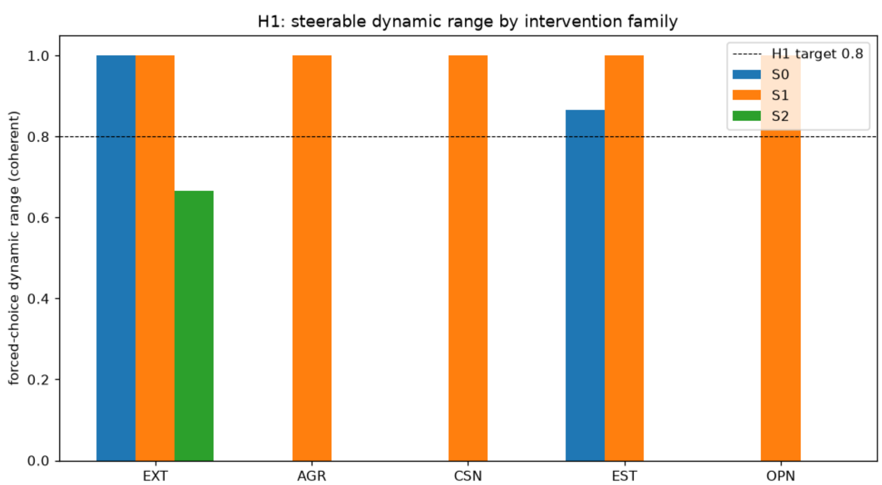

Figure 2: H1 dynamic range by family. S1 (norm-scaled, multi-layer, all-position) reaches the H1 target for every trait; S0 collapses for AGR/CSN/OPN; S2 (capping) is largely inert (below).

H2. The probe-optimal method is M2 for all traits; the steering-optimal family is S1 (all but EXT, where S0 ties). The best *readout* direction is not delivered by the intervention that steers best — the probe/steering dissociation H2 predicts. S2 capping was largely inert: its thresholds, calibrated on the character PROMPT-MEAN projection distribution, do not transfer to generation-token activations. This is a calibration finding (reported, not re-engineered), not evidence against capping in general.

Likert re-administration. Re-administering the full IPIP with no persona under steering gives a graded (10–50) view (Table 3). Downward steering lowers scores — strongest for Agreeableness (34.9→18.9); the high pole stays near baseline because baseline is already saturated. The forced-choice swing (1.0→0) is larger than the Likert swing because forced-choice is a relative ranking of items whereas Likert is absolute per-item agreement.

Table 3: Likert (10–50) re-administration, no persona, under the selected poles.

Trait	Low pole	Baseline	High pole
EXT	31.7	34.8	34.3
AGR	18.9	34.9	27.9
CSN	30.2	34.1	31.8
EST	29.1	33.2	33.6
OPN	29.7	34.5	31.7

3.3 Specificity: H5

Steering trait i (both poles) and measuring the forced-choice swing of all five traits yields the 5×5 matrix in Fig. 3. Specificity is trait-dependent (Table 4): clean for Agreeableness (3.4×) and Emotional Stability (14×), but leaky for Extraversion (1.8×), Conscientiousness (1.7×), and especially Openness, where steering moves Emotional Stability (+0.80) and Conscientiousness (+0.60) *more* than Openness itself (+0.40). **H5 is not confirmed** (2/5 pass). This is a striking dissociation: the directions are near-orthogonal *representationally* ($|\cos| < 0.12$) yet leak *causally*.

Table 4: H5: on-diagonal vs largest off-diagonal forced-choice swing. Pass = on-diagonal ≥ 2× largest off-diagonal.

Steered	On-diag	Max off-diag	Ratio	Pass
EXT	0.68	0.38	1.8×	no
AGR	0.90	0.27	3.4×	yes
CSN	1.00	0.60	1.7×	no
EST	0.93	0.07	14×	yes
OPN	0.40	0.80	0.5×	no

3.4 The Assistant Axis in Big Five coordinates: H3

At the pre-registered summary layer (L40, the Assistant-Axis target layer), the Assistant Axis is nearly *orthogonal* to every individual Big Five direction ($|\cos| < 0.07$; Table 5). The default-Assistant fingerprint is mildly high on Agreeableness (z=+0.31) but not on Conscientiousness (−0.04) and strongly *low* on Openness (−0.60), so the H3 prediction (high AGR/CSN/EST) is not met. Regressing role Assistant-Axis projection on the five Big Five coordinates over the 274 roles gives R^2=0.47 — below the 0.5 bar — leaving a 53% **residual "AI-ness"** orthogonal to human personality. **H3 is not confirmed**, and the failure is informative.

The layer profile (Fig. 4) is essential to reading this correctly: R^2 is 0.84 at layer 0 but that is an embedding/token-surface artifact (role names); it dips to 0.24–0.47 across the *functional* middle layers (L28–40), where the Assistant Axis operates, and recovers only to ∼0.6 late. We did not report the L0 value as the headline nor tune the summary layer to force a pass. The single largest

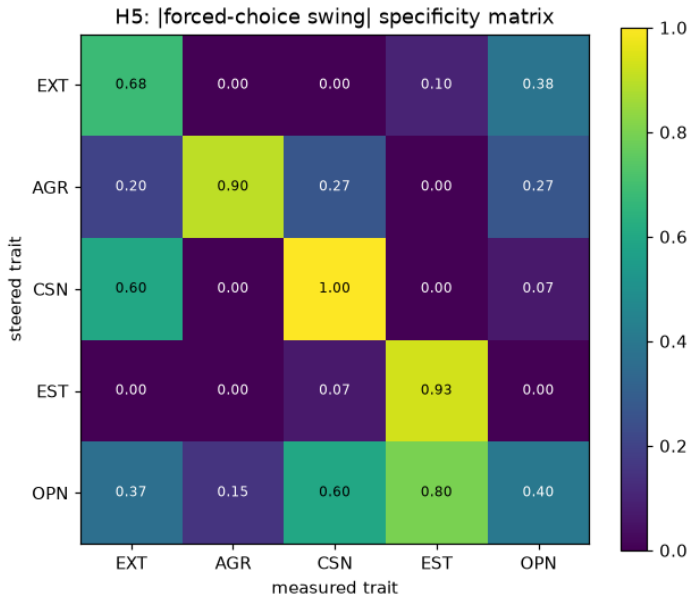

Figure 3: H5 specificity matrix: |forced-choice swing| when steering the row trait and measuring the column trait. A perfectly specific basis would be diagonal.

Table 5: H3 at L40. cos(AA, trait); Assistant fingerprint (z vs 274 roles); standardised regression β of role AA-projection on Big Five.

	EXT	AGR	CSN	EST	OPN
$\cos(\mathrm{AA}, \hat{w})$	+0.009	+0.029	−0.003	+0.003	−0.065
Fingerprint z	+0.14	+0.31	−0.04	+0.05	−0.60
Regression β	+0.26	+0.18	+0.46	+0.18	−0.56

positive residual role is, tellingly, `assistant` itself (+1.6): the Assistant persona sits far above where its Big Five profile predicts, which is precisely the "AI-ness" the residual captures. Role Big Five profiles are face-valid (e.g. most-conscientious roles: `perfectionist`, `soldier`, `robot`; most-open: `eldritch`, `witch`, `genie`).

3.5 The causal tie: H4

Steering each Big Five direction and measuring the induced change in Assistant-Axis projection (Fig. 5, Table 6) reveals a clean, interpretable causal effect for **Agreeableness**: lowering it drives the model's Assistant-Axis projection from +1.63 to −0.10 — pushing it *off* the Assistant persona — a swing of +1.23. Openness is moderate (+0.68); the other three are weak (< 0.25). The reverse direction (steering the Assistant Axis and measuring Big Five) was *inconclusive*: strong Assistant-Axis steering degraded the forced-choice into incoherence, itself consistent with the axis governing task-following. We mark H4 **partial**: the §6.3 cross-steering causal tie is present and one-directional; the pre-registered §6.2 multi-turn drift regression was not run (see Limitations).

Correlational ≠ causal. The causal picture and the H3 correlational decomposition *disagree* on which trait matters most. The regression loads Conscientiousness (+0.46) and negative Openness

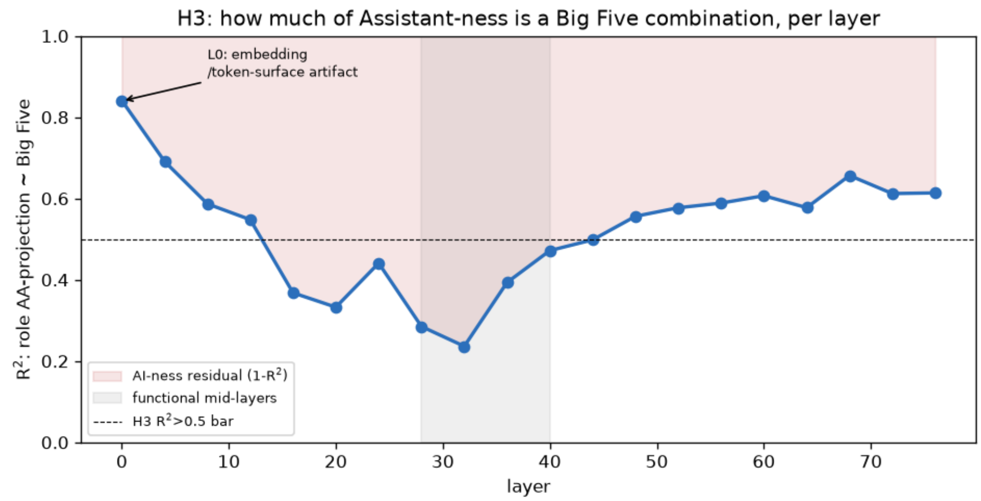

Figure 4: H3: fraction of role Assistant-ness explained by Big Five, per layer. In the functional mid-layers, over half of Assistant-ness is orthogonal to human personality.

Table 6: H4 (direction A): steering each Big Five trait, mean Assistant-Axis projection at each pole and the high−low swing.

Steered trait	AA (base)	AA (high)	AA (low)	Swing
EXT	+1.63	+1.56	+1.54	+0.02
AGR	+1.63	+1.13	−0.10	**+1.23**
CSN	+1.63	+1.55	+1.35	+0.20
EST	+1.63	+1.56	+1.34	+0.22
OPN	+1.63	+1.79	+1.11	+0.68

(−0.56) highest, with Agreeableness only +0.18; yet the causal lever is overwhelmingly Agreeableness. This is exactly why the protocol demanded both analyses: the directions that best *describe* variation across roles are not the directions that most *move* the model. Lowering agreeableness — making the model disagreeable — is what actually breaks the Assistant persona.

4 Discussion

Three dissociations organise the findings. **(1) Probing vs steering:** the directions probe almost perfectly ($\rho \approx 0.95$) yet require a strong, multi-layer, all-position intervention to steer; the naive additive use of the same direction degenerates (H1/H2). **(2) Representational vs causal specificity:** the five directions are near-orthogonal in activation space ($|\cos| < 0.12$) yet steering them produces cross-trait behavioural leakage, severely so for Openness (H5). **(3) Correlational vs causal structure:** Big Five explains under half of Assistant-ness and the Assistant Axis is geometrically orthogonal to the trait directions (H3), yet a single trait — Agreeableness — is a strong causal lever on Assistant-ness (H4). Together these argue that the "helpful Assistant" is not a point in Big Five space but a largely orthogonal attractor, anchored behaviourally by agreeableness and otherwise irreducible to human personality structure.

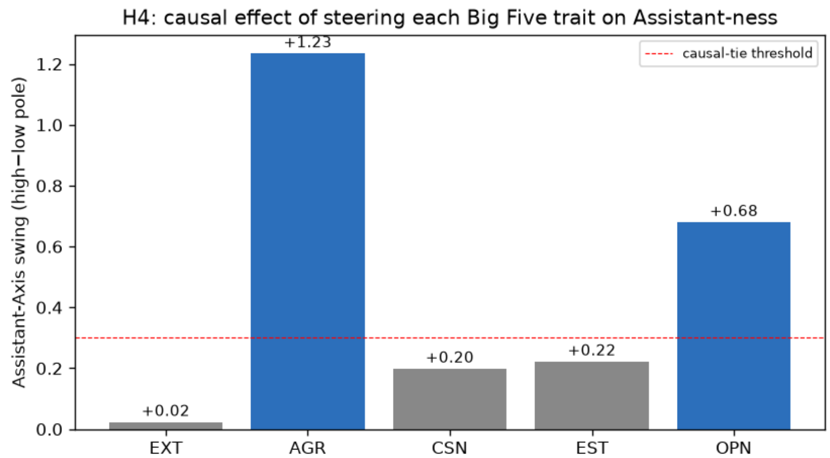

Figure 5: H4 causal map: effect of steering each Big Five trait on Assistant-ness. Agreeableness is the dominant causal lever.

5 Deviations and limitations

- **Stimuli.** The designated source repo is unavailable; all stimuli were harvested from the paper (the 406-character parse is self-checking), with the adjective set substituted by the Saucier Mini-Markers.
- **Positions.** Only the two prompt-side readouts were collected; GEN-MEAN was skipped (never selected; reversible).
- **Norm reference** for S1 was measured on neutral Alpaca responses, not LMSYS-Chat-1M.
- **Split** uses summed-z quintile stratification, not the literal joint 5-trait quintile (which drains the test set to $\sim$11 characters).
- **S2 capping** thresholds were derived per protocol from the character projection distribution and proved inert at generation time; not re-tuned.
- **Confirmatory metrics** (Likert, specificity, open-ended) were run on the selected config per trait, not the full grid.
- **H4 §6.2** (multi-turn drift regression over $\geq$200 conversations with an auditor) was not run; the stronger §6.3 cross-steering causal tie is reported instead. This is the single remaining pre-registered item.
- **Judge validation** against human labels was not possible (no human raters); the judge model is used as-is.
- Index files are JSON rather than parquet (pandas/pyarrow absent).

6 Conclusion

We built Big Five directions for `Llama-3.3-70B` that probe near-perfectly and steer fully under a strong intervention (H1/H2), with trait-dependent causal specificity (H5). Expressing the Assistant

Axis in this basis shows that Assistant-ness is largely orthogonal to human personality — under half explained by Big Five, with the Assistant role its own largest residual (H3) — and yet is causally anchored by Agreeableness (H4). The recurring theme is that *geometry, correlation, and causation come apart*: a direction can describe without moving, and orthogonality in representation does not guarantee independence under intervention.

Artifacts. All results are in `results/bigfive/llama-3.3-70b/`: `character_profiles.json`, `bigfive_basis.npz`, `direction_bank.npz`, `stage1_selection.json`, `steering_grid.json`, `steering_results.j` `stage3_decomposition.json`, `cross_steering_matrix.json`, and figures. Code is in `src/bigfive/`.